

Course faculty:

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Course assistants:

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Amount of ECTS credits: 4 credits

Class hours: an introductory lecture and 19 problem-solving sessions (40 class hours).

Background and course rationale:

This course provides an introduction to Probability Theory, which is a building block and a prerequisite for many subsequent courses, both mathematical and economical ones. Besides providing sufficient background, the course will enable students to make better everyday decisions in situations involving a certain degree of uncertainty.

Prerequisites: None

Learning outcomes:

Probability essentials

2024/2025, Autumn

By the end of this course students will be able to:

- interpret logical propositions
- formulate propositions correctly and recognize logical mistakes in common situations, find value of propositions analytically
- construct negations to propositions
- define and identify basic set operations
- solve basic combinatorial problems

- construct sample spaces
- identify a priori and a posteriori probabilities of various events in discrete sample spaces
- find expected values and other characteristics of discrete random variables.

How we will work at this course (what kinds of tasks and activities students are expected to do):

This 8-week course consists of 20 classes of topic discussion, problem-solving, and revision. Attending classes on a regular basis is highly recommended.

The course is organized according to the “flipped learning” principle. It means that students must process some sources or tasks on his/her own before attending class. You can find a detailed script for every lesson on Moodle. This principle is designed to maximize efficiency of time the instructor and students spend together.

Students can best succeed in the course if they work systematically and intensively, use a proactive approach and effective study methods, show interpersonal, time-management, and active-learning skills.

Students will take 4 tests during the course. Apart from tests, homework is also considered as an assessment activity. Although doing homework and Khan Academy assignments usually helps to pass the tests, it may not be enough to get the highest marks, which typically requires advanced problem-solving skills and some creative powers. Additionally, during practice sessions, there will be a short quiz each session to check students' preparation based on the lecture material. *If a student misses a quiz or quizzes for a valid reason, then he/she will have the opportunity to write them at the end of the course, but no more than three of them.*

If students have any questions, which is entirely natural while studying, the good strategy is to ask them immediately during classes or office hours.

Work languages on this course are English and Ukrainian.

Grading Policy

No	Assessment activities	The number of activities	Points per activity	The total number of points
1	Test 1	1	15	15
2	Test 2	1	15	15
3	Test 3	1	15	15
4	Test 4	1	8	8
5	Quizzes	16	1.5	24
6	Homeworks	4	4	16
7	Khan Academy exercises	3	4	12
9	Project	1	0	0
	TOTAL			105

To complete the course students should get at least 60 points.

Taking into account the fact that the course is about mathematics and studying at the university involves observing the principle of scientificity, all answers and conclusions should be justified. For the same reason, you can only use formulas whose derivation is familiar to you.

Simultaneously a detailed explanation of the process of solving the problem is a good practice that allows a student to appeal in case he/she does not agree with evaluation of his/her work.

Assignments should be graded in 7 days after the deadline. Received grades can be appealed in 3 days after getting a result. If there is no appeal in 3 days, the grade becomes final. Appealing may result in both a higher or a lower grade.

Written assignments, that have no strict time restriction, must be completed with the following requirements:

- each task must be visually separated from others;

- tasks should go in the same order as in assignment (or must be accompanied by list of contents);
- answers may be highlighted;
- if the assignment is submitted online your work must be submitted in .pdf format, filename should contain first and last name of executant. It is also recommended to add name of assignment to filename;
- photos should be rotated by 0 degrees;
- photos should be clearly visible.

Work should be submitted a day before the deadline, and the last day is reserved only for force majeure situations (like power outages). Therefore, force majeure can only be used as a justification if the force majeure circumstances last more than a day. Submissions after the deadline will not be graded.

If the assignment has strict time restrictions it is recommended to follow the principles of mutual respect and common sense.

Tests:

The tests will consist of several tasks. You can prepare a brief theoretical summary on an A4-size sheet without including specific examples, and use it during the test. The summary should be submitted together with your work. You can't use any other sources during the tests.

Reading List:

[1] Richmond B., Richmond T. A Discrete Transition to Advanced Mathematics. Amer. Math. Soc., 2009.

[2] Soo T. Tan. Applied Mathematics for the Managerial, Life, and Social Sciences, 7th Edition. Cole/Brooks, 2014.

[3] Wazir I., Garry T., Nakamoto J., Frederick K., Lumb S. Mathematics Applications and Interpretation for the IB Diploma. Higher Level. Pearson, 2019.

[4] <https://www.khanacademy.org/math/statistics-probability>

Course venue: KSE Dragon Capital Building, 3 Mykola Shpak Street, Kyiv

Academic integrity warning

Academic integrity is highly valued by the KSE community and its vast majority of students. We do not tolerate plagiarism and cheating. Very competitive essence of the

program calls for absolutely equal treatment of all students. Even a single case of violation of academic integrity is a serious misdemeanor which may lead to unjust redistribution of grades and, consequently, overall rankings, possible grants, fee reductions and other merit-based awards. The minimum penalty for a student caught cheating or plagiarizing during a quiz, a midterm or the final exam will be a zero for this assignment with the consecutive decrease of the final course grade by at least one letter-grade. Repetitive violation of academic integrity may lead to the dismissal of the student from KSE, upon approval of the faculty committee. Please take this issue seriously and responsibly.

Course Plan

Week 1

Introductory lecture for all students of the course

Block 1. Basics of logic and set theory

Preparation lecture materials 1. Logical propositions, true/false propositions.

Practice session 1. Logical propositions, true/false propositions.

Preparation lecture materials 2. Logical operations, truth tables.

Practice session 2. Logical operations, truth tables.

Week 2

Preparation lecture materials 3. Building negations to claims.

Practice session 3. Building negations to claims.

Preparation lecture materials 4. Sets and subsets, set operations, Euler diagrams.

Practice session 4. Sets and subsets, set operations, Euler diagrams.

Week 3

Preparation lecture materials 5. The Number of Elements in a Finite Set.

Practice session 5. The Number of Elements in a Finite Set.

Homework 1

Revision session before Test 1

Block 2. Set theory and combinatorics

Preparation lecture materials 6. Cartesian product, multiplication and addition rules.

Practice session 6. Cartesian product, multiplication and addition rules.

Week 4

Test 1

Preparation lecture materials 7. Permutations and Combinations.

Practice session 7. Permutations and Combinations.

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Block 3. Probability, rules of probability

Preparation lecture materials 8. Event space, events.

Practice session 8. Event space, events.

Week 5

Homework 2

Preparation lecture materials 9. Probability definition. Rules of probability.

Practice session 9. Probability definition. Rules of probability.

Preparation lecture materials 10. Usage of combinatorics in probability word problems

Practice session 10. Usage of combinatorics in probability word problems.

Revision session before Test 2

Week 6

Test 2

Block 4. Conditional probability, independent events, Bayes' rule

Preparation lecture materials 11. Conditional probability and independent events.

Practice session 11. Conditional probability and independent events.

Preparation lecture materials 12. Total probability law (chain rule).

Practice session 12. Total probability law (chain rule).

Homework 3

Week 7

Preparation lecture materials 13. A posteriori probability, Bayes' rule.

Practice session 13. A posteriori probability, Bayes' rule.

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Preparation lecture materials 14. A posteriori probability, Bayes' rule.

Practice session 14. A posteriori probability, Bayes' rule.

Revision session before Test 3

Homework 4

Week 8

Test 3

Block 5. Random variables. Random distributions and their characteristics

Preparation lecture materials 15. Random distributions. Expected value.

Practice session 15. Random distributions. Expected value.

Preparation lecture materials 16. Random distributions. Variance and standard deviation.

Practice session 16. Random distributions. Variance and standard deviation.

Test 4

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Project